DJANGO WEB APPLICATION • TASK 3

MVT ARCHITECTURE

# Shopping Hub Portal

Multi-Marketplace Navigation & Video Showcase Interface

| PROJECT NAME | APP NAME | FRAMEWORK | STATUS |
| --- | --- | --- | --- |
| **task3** | **shoppingapp** | **Django 6.1 / Python 3.12** | **Verified & Complete** |

**Executive Summary:** Task 3 demonstrates core Django fundamentals: Model-View-Template (MVT) pattern, dynamic static media routing, and responsive frontend integration. The application serves a sleek centralized shopping hub featuring marketplace cards for Amazon, Flipkart, and Meesho alongside an embedded media player.

## 1. Data Models (`shoppingapp/models.py`)

The `models.py` defines the data persistence tier. In this frontend-focused project, standard Django DB model inheritance is imported ready for future inventory schema extensions.

```python
from django.db import models

# Create your models here.
```

## 2. URL Routing (`task3/urls.py`)

The root URL configuration maps HTTP incoming requests to view controllers and enables static file serving for assets.

| URL Pattern | View Function | Route Name | Description |
|---|---|---|---|
| `''` (Root) | `views.shoppingapp` | `'home'` | Default landing view presenting the Shopping Hub |
| `'shoppingapp/'` | `views.shoppingapp` | `'shoppingapp'` | Direct app namespace endpoint |
| `'admin/'` | `admin.site.urls` | `'admin'` | Django Administration control panel |

```python
"""
URL configuration for task3 project.

The `urlpatterns` list routes URLs to views. For more information please see:
    https://docs.djangoproject.com/en/6.1/topics/http/urls/
Examples:
Function views
    1. Add an import:  from my_app import views
    2. Add a URL to urlpatterns:  path('', views.home, name='home')
Class-based views
    1. Add an import:  from other_app.views import Home
    2. Add a URL to urlpatterns:  path('', Home.as_view(), name='home')
Including another URLconf
    1. Import the include() function: from django.urls import include, path
    2. Add a URL to urlpatterns:  path('blog/', include('blog.urls'))
"""
from django.contrib import admin
from django.urls import include, path
from shoppingapp import views

urlpatterns = [
    path('', views.shoppingapp, name='home'),
    path('shoppingapp/', views.shoppingapp, name='shoppingapp'),
    path('admin/', admin.site.urls),
]

from django.contrib.staticfiles.urls import staticfiles_urlpatterns
urlpatterns += staticfiles_urlpatterns()
```

## 3. View Controller Logic (`shoppingapp/views.py`)

The controller handles incoming HTTP requests and dispatches the rendered HTML template response.

> **Key Functionality:**
>
> - `shoppingapp(request)`: Renders `shopping.html` template into an HTTP response using Django's template engine.
> - Static asset bindings for stylesheets and e-commerce marketplace logos.

```python
from django.shortcuts import render
from django.http import HttpResponse
from django.template import loader

# Create your views here.


def shoppingapp(request):
    return render(request,'shopping.html')


```

## 4. Live Application Output Screenshot

Below is the live rendered output captured from the local server instance (`http://127.0.0.1:8001/`), displaying responsive marketplace tiles and the video player showcase.

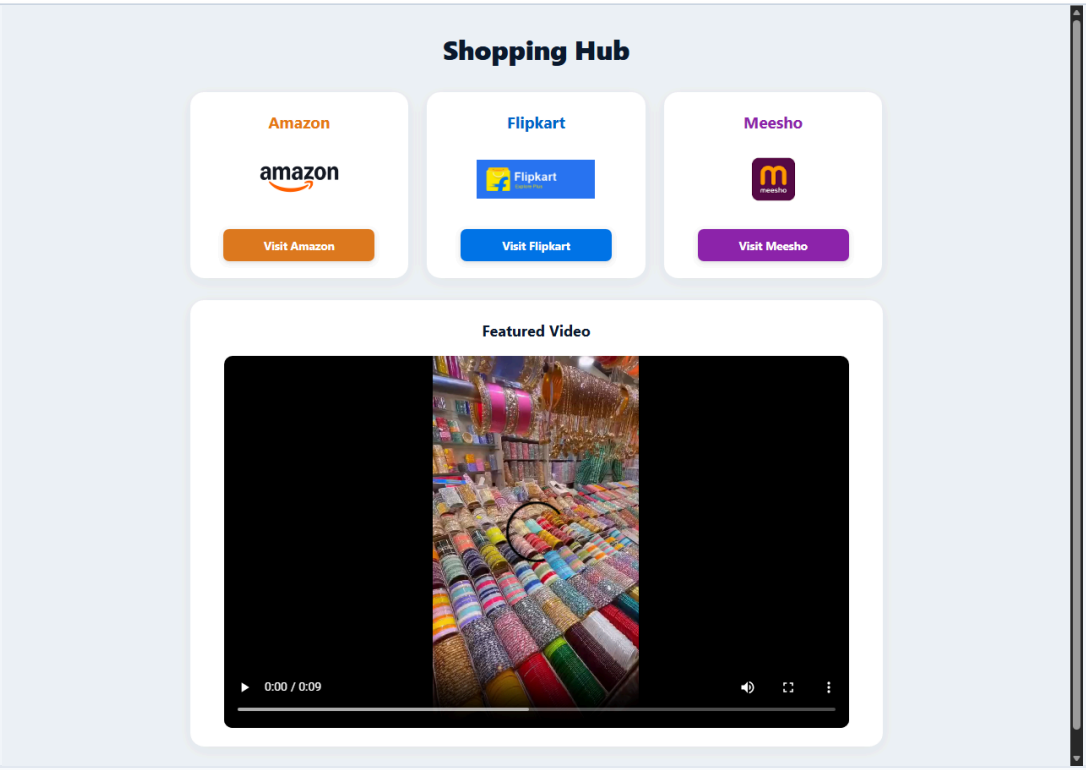



**Figure 1.1:** Shopping Hub User Interface featuring interactive cards for Amazon, Flipkart, Meesho, and the HTML5 media showcase container.